B S P Krishna Chaitanya Reddy

ai23btech11002@iith.ac.in | chaitanya657657@gmail.com | +91 9121720020 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.35
NARAYANA JUNIOR COLLEGE	Class XII (TSBIE)	2023	97.00%
KENNEDY EM HS	Class X (BSEAP)	2021	99.70%

SCHOLASTIC ACHIEVEMENTS

- **NSEC:** Certificate of Merit for ranking in the **top 1% of 2,827 candidates** and securing **State Topper**.

PROJECTS

Loan Default Prediction using Machine Learning | Business Analytics

- Analyzed **2.26M LendingClub loan records** using Python, performing data cleaning, feature engineering, **EDA, correlation analysis, VIF, and Lasso selection**, reducing 151 raw variables to **77 actionable features**.
- Compared **Logistic Regression, LDA, Decision Tree, Random Forest, XGBoost, and SVM**, with **XGBoost achieving 0.733 AUC**; threshold tuning from **0.50 to 0.23** raised default recall from **10.8% to 62.0%**.
- Built **portfolio risk tiers and scored 912K active loans**, identifying **32.1% as high-risk**; analyzed feature importance and borrower segments to derive actionable insights for **risk monitoring, underwriting, and portfolio management**.

Bharat Taxi Customer Analytics | Text Analytics

- Analyzed **2,325 customer reviews, comments, and news records** across Google Play, App Store, YouTube, and Google News to identify **customer sentiment, product pain points, and service trends**.
- Applied **VADER sentiment analysis, TF-IDF, keyword co-occurrence networks, LDA topic modeling, and NER** to identify key themes around **app functionality, pricing, waiting time, drivers, and competition**.
- Combined **topic-level sentiment and platform-wise analysis** to identify recurring customer issues such as **ride cancellations, delays, pricing, and customer support**, translating findings into product and service improvement recommendations.

Bayesian Statistical Analysis of Meteorological Data | Bayesian Statistics

- Analyzed **hourly meteorological data** using Bayesian inference, modeling **temperature, cloud cover, precipitation, radiation, and wind speed** with conjugate and non-conjugate probabilistic models.
- Applied **Bayes factors, posterior estimation, grid approximation, and Metropolis-Hastings sampling** to quantify uncertainty, compare hypotheses, and estimate meteorological parameters.
- Built a **Bayesian multiple linear regression model using Gibbs sampling**, identifying the relationship between **daytime temperature, humidity, and solar radiation** with quantified posterior uncertainty.

ML Fundamentals | Foundations of Machine Learning

- Built **KNN and regression models from scratch in Python**, analyzing **neighborhood selection, error surfaces, overfitting, regularization, kernel methods, and SGD-based optimization** on real-world datasets.
- Implemented and compared **Least Squares, LDA, FDA, and Perceptron classifiers** on synthetic datasets, visualizing **decision boundaries** and evaluating classification performance across different learning approaches.
- Developed a **feed-forward neural network from scratch using NumPy** for **MNIST digit classification**, experimenting with **activation functions, optimizers, dropout, and regularization** to improve generalization and reduce overfitting.

SKILLS

- **Languages:** C,C++, Python, SQL, Assembly
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, OpenCV, Scikit-learn, PyTorch, FastAPI
- **Tools & Technologies:** Git, RAG, Qdrant, MCP, llama-cpp, Jupyter Notebook, Google Colab, LaTex, Solid Edge, WSL

RELEVANT COURSES

Business & Analytics: Business Analytics, Text Analytics, Predictive Modeling

Mathematics: Applied Statistics, Bayesian Statistics, Probability & Random Variables, Calculus, Matrix Theory

Artificial Intelligence: Foundations of Machine Learning, Deep Learning, Natural Language Processing, Computer Vision

Computer Science: DBMS, Data Structures, Algorithms, Introduction to Programming, Discrete Mathematics

POSITIONS OF RESPONSIBILITY

Head of Events -- PRAYAS Club, IITH Rural Development Centre

- Led end-to-end planning and execution of club events, coordinating across different domains, managing timelines, overseeing event logistics, and ensuring smooth and timely execution from planning through completion.

Transport Coordinator -- Elan & nVision Club

- Coordinated event transportation, managing vehicle scheduling, and route planning for timely execution.

EXTRACURRICULAR

- Completed **120+ hours of community service as part of NSS,** including government-school teaching, Swachh Bharat initiatives, plantation drives, awareness campaigns, and theme-based essay writing activities.

INTERESTS

- Playing Badminton, Mathematical problem solving.